

Solution

ELECTRICITY -09

Class 10 - Science

Section A

1. The household appliances should be connected in parallel in order to get equal voltage for each appliance and ensure that if one switch is 'on' or 'off', others are not affected.

2. Effective resistance in series combination, $R_s = R_1 + R_2 + R_3 + R_4 = (3 + 4 + 5 + 12) = 24\Omega$.

Current remains same in series combination.

Here, $V = 12V$.

We know that, $V = IR$

Therefore, $I = \frac{V}{R_s} = \frac{12V}{24\Omega} = 0.5A$.

Current flowing through 12Ω resistor = $0.5A$

3. We will connect the ammeter or voltmeter arbitrary in the circuit and observe the deflection of the pointer. If the pointer strikes the stopper (i.e., before the zero reading of the scale) or deflection occurs in the opposite direction, then by interchanging the terminal connections, we can use these devices properly in the circuit.

4. Least count of voltmeter = $0.05V$

Observed reading of voltmeter = $15 \times 0.05 = 0.75V$.

5. Equivalent resistance in series (R_s) = $R + R = 2R$

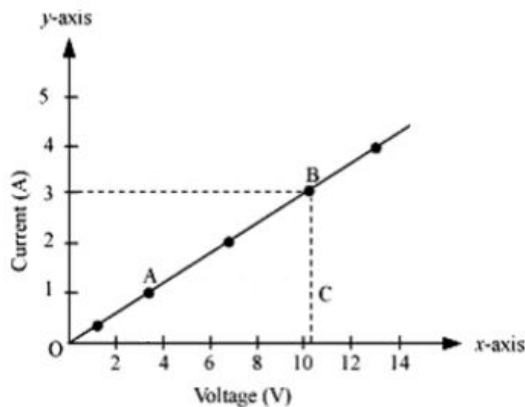
Equivalent resistance in parallel (R_p) = $\frac{1}{R_p} = \frac{1}{R} + \frac{1}{R}$

$$\Rightarrow R_p = \frac{R}{2}$$

Hence, Difference in equivalent resistance of series combination and parallel combination is = $R_s - R_p = 2R -$

$$\frac{R}{2} = \frac{4R - R}{2} = \frac{3R}{2}$$

6. The plot of Potential difference V , taken on x-axis versus Current I , taken on y-axis flowing through the circuit is given below:
Graph:



We know that, $V = IR$

The slope of the curve gives resistance (R)

So, Resistance of resistor, $R = \frac{AC}{AB} = \frac{10.2-3.4}{3-1} = \frac{6.8}{2} = 3.4\Omega$

7. If the deflection on ammeter (or voltmeter) scale goes beyond the full scale, we can infer that

i. the higher range ammeter must be used for measuring the higher value of current in the circuit.

ii. the applied voltage will be very high so there will be a need of higher range voltmeter to measure the applied voltage.

If deflection takes place in opposite direction, the device is not properly connected in the circuit. We must interchange the terminal connection, so that devices can be used properly in the circuit.

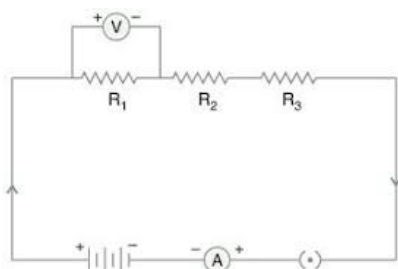
8. As the two equal conductors are connected in series the resultant resistance of the combination too doubles.

Heat Produced in the coil = i^2Rt ,

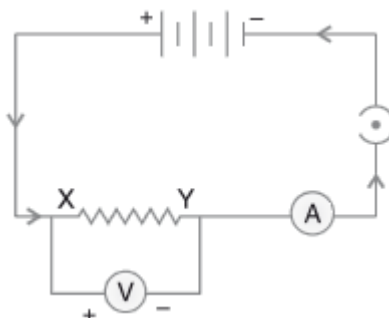
If the electric current and the time period are same the heat produced by the coil is directly proportional to the resistance hence it is directly proportional to the length.

So if the length of the conducting wire is doubled without changing its area of cross section the heat produced by it is also doubled.

9. Charges can move if there is a difference of potential difference along the conductor. Electric cell or a battery consisting of one or more cells is used to maintain the potential difference in a circuit
10. The resistivity of manganin wire is $44 \times 10^{-8} \Omega\text{m}$ while that of nichrome wire is $100 \times 10^{-8} \Omega\text{m}$.
So, manganin wire offers less resistance as compared to the nichrome wire for the same length and same area of cross-section.
Hence, the value of current through the manganin wire will increase and the potential difference across its ends will decrease.
11. Circuit diagram to show R_1 , R_2 and R_3 connected in series with a battery, ammeter and a key. Voltmeter V is connected parallel to R_1 :



12. The range of ammeter = 3 A
Number of divisions in ammeter = 30
Least count = $\frac{\text{Range}}{\text{Total Division}} = \frac{3}{30} = 0.1 \text{ A}$
13. According to the question, Resistance of each resistor = 2Ω
We know that, the resistance are added if they are connected in series combination.
As, $R_{\text{effective}} = (2 + 2 + 2) = 6\Omega$.
Therefore, All the given set of resistors must be connected in series, so that the equivalent resistance is 6Ω .
Reason: The effective length of all the resistors in the series combination increases. Hence, the equivalent resistance of the combination increases as $R \propto l$.
14. We always connect an ammeter (a low resistance device) in series and voltmeter (a high resistance device) in parallel while performing the experiment. It is shown below in circuit diagram:



15. The given resistor is cut into five equal parts.
Therefore, Resistance of each part = $\frac{R}{5}$
When these resistors are connected in parallel,
Then the effective resistance is
 $\frac{1}{R'} = \frac{5}{R} + \frac{5}{R} + \frac{5}{R} + \frac{5}{R} + \frac{5}{R} = \frac{25}{R}$
 $\Rightarrow \frac{1}{R'} = \frac{25}{R}$
 $\Rightarrow \frac{R}{R'} = 25$

Section B

16. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
17. (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
18. (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
19. (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

Section C

20. (a) (i) Resistivity - Since the Resistivity is a property of a substance, hence it remains the same for both the wires.
(ii) Resistance - As both the wires are of different cross sectional areas, so both wires are considered as different objects and will have different resistances.
(b) (i) $\because R = \frac{\rho L}{A}$
For wire A (R_1) = $\frac{\rho l}{A_1}$

For wire B (R_2) = $\frac{\rho l}{A_2}$

$\frac{R_2}{R_1} = \frac{A_1}{A_2}$ (∵ Resistance is inversely proportional to the area of cross section of the conductor.)

$$\therefore R_1 = 4R_2$$

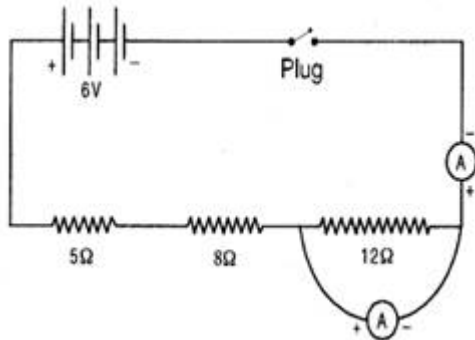
$$\frac{A_1}{A_2} = 1 : 4$$

$$(ii) \left(\frac{r_1}{r_2}\right)^2 = \frac{1}{4} \quad (\because \text{Ratio of the radii of wire is square root the ratio of the area})$$

$$r_1 : r_2 = 1 : 2$$

21. Modified circuit is as shown. Since 5Ω , 8Ω and 12Ω are in series, therefore the total resistance in series.

$$R_s = R_1 + R_2 + R_3 = 5 + 8 + 12 = 25\Omega$$



$$\text{Current through circuit } I = \frac{V}{R} = \frac{6}{25} = 0.24A$$

$$\therefore \text{Reading of ammeter} = 0.24A$$

$$= I.R. = 0.24 \times 12$$

$$= 2.88\Omega$$

22. a. Ohm's law states that the current through a conductor between two points is directly proportional to the potential difference across the two points and inversely proportional to the resistance between two points.

b. The slope of the displacement time graph gives us the physical quantity called velocity.

The SI unit of velocity is metre per second (m/s).

c. 1 KWh is defined as the 1000 J energy is used in 1 hour.

$$\text{Mathematically, } 1 \text{ KWh} = 3.6 \times 10^6 \text{ J/s.}$$

23. a. Both resistance and bulb are connected in series, therefore, total resistance:

$$R = R_1 + R_2$$

$$= 4 + 20 = 24 \Omega$$

$$\text{b. Current through circuit, } I = \frac{V}{R} = \frac{6}{24} = 0.25 \text{ A}$$

c. (i) P.D. across electric lamp,

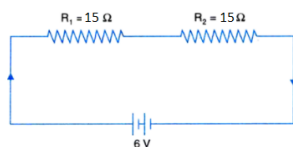
$$V_2 = IR_2 = 0.25 \times 20 = 5V$$

(ii) P.D. across Conductor,

$$V_1 = IR_1 = 0.25 \times 4 = 1V$$

$$\text{d. Power of lamp} = V_2 I = 5 \times 0.25 = 1.25 \text{ W}$$

24. i. Series combination:



$$\text{Here, } R_1 = 15\Omega, R_2 = 15\Omega$$

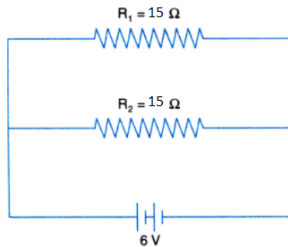
$$\text{Total resistance of the series combination is given by, } R_s = R_1 + R_2 = 15 + 15 = 30\Omega$$

$$\text{Applied voltage is given by } V = 6V$$

$$\text{So, Power consume by resistor given by } P_1 = \frac{V^2}{R_s} = \frac{6^2}{30} = \frac{36}{30} = 1.2 \text{ W}$$

$$\text{Thus, power consume in series is } 1.2W$$

ii. Parallel combination:



Here, $R_1 = 15\Omega$, $R_2 = 15\Omega$

Total resistance R_p of the parallel combination is given by,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{15} + \frac{1}{15} = \frac{2}{15}$$

$$\text{Power, } P_2 = \frac{V^2}{R_p} = \frac{6^2}{15} = \frac{36 \times 2}{15} = 4.8 \text{ W}$$

Power consume in parallel is 4.8W

Now, the ratio of the powers consumed in the two combinations is given by,

$$\frac{P_1}{P_2} = \frac{1.2}{4.8} = \frac{12}{48} = \frac{1}{4}$$

$$\therefore P_1 : P_2 = 1 : 4$$

25. Electric energy is the total amount of work done by the current in a given time or electric energy is the total amount of energy consumed in an electric circuit in a given time. The total energy consumed not only depends upon the power of the appliance but also upon the time for which the power is maintained.

If a power P (watt) is maintained for t second, the work done or the energy consumed $\Omega(\text{joule}) = P(\text{watt}) \times t(\text{second})$

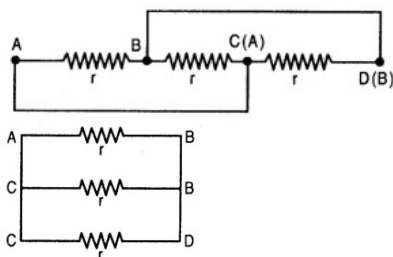
$$\text{Or } W = P \left(\frac{J}{s} \right) \times t(s) = Pt \text{ joule}$$

But $P = VI \therefore \Omega = VIt$ joule

Practical Unit of Electric Energy. The usual unit of electric energy i.e. joule is too small unit. Hence in practice, a bigger unit of electric energy called kWh (kilowatt-hour) is usually used. This unit is also called B.O.T. (Board of Trade Unit). In all the English speaking countries including India; kWh or B.O.T. is the practical or commercial unit of electric energy. The cost of electric energy consumed is rated in terms of the B.O.T. unit.

The unit kWh is equal to the work done or energy consumed when a power of 1kW is consumed for 1 hour.

26. Reducing the actual circuit to an equivalent circuit i.e. we find that the three resistors, each equal to r , are just placed parallel to each.

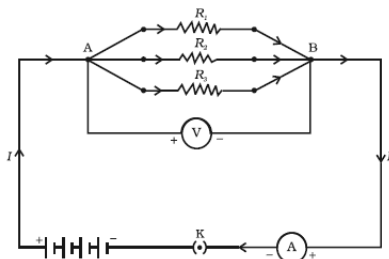


$\therefore$ Equivalent resistance R_p is given by

$$\frac{1}{R_p} = \frac{1}{r} + \frac{1}{r} + \frac{1}{r} = \frac{3}{r} \text{ or } R_p = \frac{r}{3}$$

27. a. Total current $I = I_1 + I_2 + I_3$

Let R_p be the equivalent resistance of R_1, R_2, R_3 . Then the total current $I = \frac{V}{R_p}$



On applying ohm's law for each R_1, R_2, R_3

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$$

$$\therefore I = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) = \frac{V}{R_p}$$

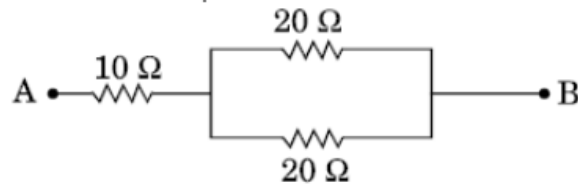
$$\therefore \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\text{b. } R_1 = 20\Omega \text{ and } R_2 = 20\Omega$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_p} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20} = \frac{1}{10}$$

$$\Rightarrow R_p = 10\Omega$$

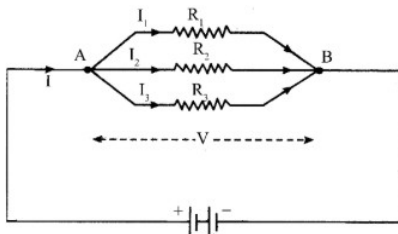


Equivalent resistance of the network = $R_{eq} = R_s + R_p = 10 + 10 = 20\Omega$

Hence, the equivalent resistance of the given network is 20Ω

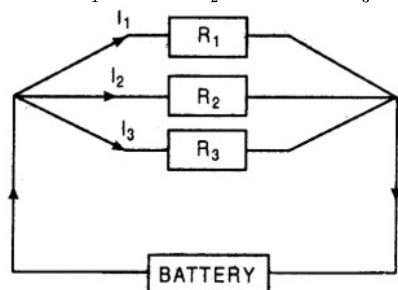
28. A circuit in which two or more resistors are connected across common points so as to provide separate paths is called parallel circuit.

In this case, the same potential difference will be maintained between the two ends of every resistor and the current will divide itself in various branches.



Let the resistors R_1 , R_2 and R_3 be joined in parallel to the points A and B. Let the current I reaching A divide itself into three parts I_1 , I_2 and I_3 along R_1 , R_2 and R_3 respectively. Let V be the potential difference between the points A and B. The current flowing in the individual resistors are then given by :

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2} \text{ and } I_3 = \frac{V}{R_3}$$



Let R_p be the resistance of the combination, then $I = \frac{V}{R_p}$ But $I = I_1 + I_2 + I_3$

$$\text{or } \frac{V}{R_p} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \text{ or } \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

The result holds true for any number of resistors.

29. i. **Given:**

Length of first wire $x = 30\text{ cm}$

Length of second wire $y = 10\text{ cm}$

Radius of first wire = 2 cm

Radius of second wire = 1 cm

To find :

Ratio of resistance of both wires

Solution :

Radius of first wire = 2 cm

So, Area of cross section of first wire = $\pi r^2 = \pi \times 2^2 = 4\pi\text{cm}^2$

Radius of second wire = 1 cm

So, Area of cross section of second wire = $\pi r^2 = \pi \times 1^2 = \pi\text{cm}^2$

General formula of resistance $R = \rho \frac{l}{a}$

Resistivity of both wires is same because both wires are made of same copper material.

So, Resistance of first wire = $r_1 = \rho \left(\frac{30}{4\pi} \right) \Omega$

and Resistance of second wire = $r_2 = \rho \left(\frac{10}{\pi} \right) \Omega$

So, the ratio of both resistance = $\frac{r_1}{r_2} = \frac{\rho \frac{30}{4\pi}}{\rho \frac{10}{\pi}} = \frac{30}{40} = \frac{3}{4}$

So the ratio of first and second wire = 3 : 4

ii. The resistance of the electric bulb is 2 ohms

Explanation:

Given that,

Current flowing in the circuit, $I = 500 \text{ mA} = 500 \times 10^{-3} \text{ A} = 0.5 \text{ A}$

The resistance of the conductor, $R_c = 10 \pi$

The equivalent resistance of the series combination of an electric lamp and a conductor is given by:

$$R = R_e + R_l$$

Voltage of the circuit, $V = 6 \text{ V}$

Let R is the effective resistance. It is given by:

$$R = \frac{V}{I}$$

$$R = \frac{6}{0.5}$$

$$R = 12 \text{ V}$$

In a series combination, the current flowing through each circuit is the same. The resistance of the lamp is given by:

$$R_l = R - R_e$$

$$R_l = 12 - 10$$

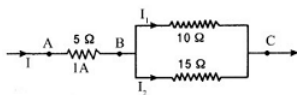
$$R_l = 2 \Omega$$

So, the resistance of the electric bulb is 2 ohms. Hence, this is the required solution.

30. Let us first find the total resistance. Now resistance of 10Ω and 15Ω are in parallel.

If R_p is the effective resistance between B and C, then

$$\frac{1}{R_p} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} \text{ Or } R_p = 6 \Omega$$



Again AB and BC are in series.

Therefore, the total resistance = Resistance between A and B plus resistance between B and C.

i.e. Total resistance = $5 \Omega + 6 \Omega = 11 \Omega$

Potential difference between A and B = $IR = 1 \times 5 = 5 \text{ V}$

Potential difference between B and C = $1 \times 6 = 6 \text{ V}$

Potential difference between A and C = $1 \times 11 = 11 \text{ V}$

Current through A and B = 1 A

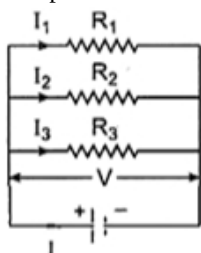
This current divides into two parts, one part I_1 passing through 10Ω and other part I_2 passing through 15Ω each producing a P.D. of 6V (between B and C)

$$\therefore I_1(10) = 6 \text{ or } I_1 = \frac{6}{10} = 0.6 \text{ A and } I_2(15) = 6 \text{ or } I_2 = \frac{6}{15} = 0.4 \text{ A}$$

31. In a parallel combination of three resistance R_1 , R_2 and R_3 , the current in each of the resistance is different. If I is the current drawn from the cell then it is divided into branches as I_1 , I_2 and I_3 . Thus,

$$I = I_1 + I_2 + I_3$$

The potential difference across each of these resistances is the same.



Thus, from Ohm's law

$$\text{current flow through } I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$$

If R is the equivalent resistance then,

$$I = \frac{V}{R}$$

$$\therefore \frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

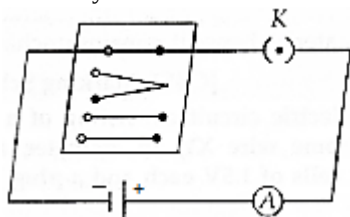
$$\text{In parallel combination } \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

32. The resistivity of substance is numerically equal to the resistance of the rod of that substance which is 1 meter long and 1 m^2 in cross-section. The S.I. unit of resistivity is ohm metre (Ωm).

If l is the length of the conductor, A is an area of the cross-section and R its total resistance then,

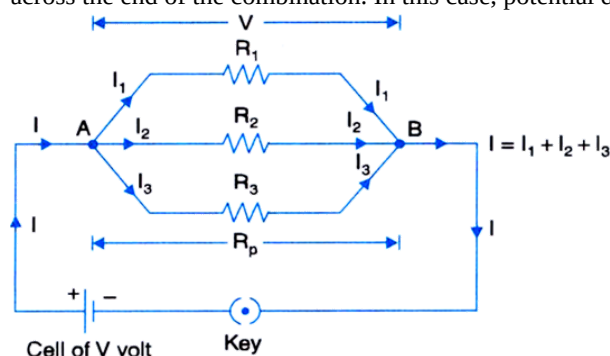
$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

Where ρ is a constant of proportionality and is called the electrical resistivity of the material of the conductor. The SI unit of resistivity is Ohm meter.



Now, plug the key. Note the current in the ammeter. Now replace the wire by a thicker nichrome wire, of the same length. The thicker wire has a larger cross-sectional area. Again note down the current through the circuit. Instead of taking a nichrome wire, connect a copper wire in the circuit. Again note down the current.

33. a. Let us consider three resistance R_1 , R_2 and R_3 are connected in parallel combination. A battery of V volts has been applied across the end of the combination. In this case, potential difference across all the resistances will be the same.



Suppose the total current flowing in the circuit is I, then the current through resistances R_1 , R_2 and R_3 will be I_1 , I_2 and I_3 respectively.

$$\text{Total current, } I = I_1 + I_2 + I_3$$

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3} \text{ (By Ohm's Law)}$$

$$\therefore \frac{V}{R_p} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

- b. $R_1 = R_2 = 12 \text{ ohms}$, $V = 6 \text{ volts}$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{12}$$

$$\therefore R_p = 6\Omega$$

$$\text{Thus current from the battery, } I = \frac{V}{R_p} = \frac{6V}{6\Omega} = 1 \text{ A}$$

34. Let a charge Q be moved from one point to the other, in the electric field and W be the work done, the potential difference V between two points is given by :

$$V = \frac{W}{Q} \text{ if } Q = +1\text{C, then } V = W$$

$\therefore$ the potential difference between two points in the electric field is defined as the amount of work done in moving a unit positive charge from one point to the other against electrostatic force due to electric field.

SI units of P.D. is volt. Potential difference between two points is said to be 1 volt if 1 J of the work is done in moving a charge of 1 C from one point to the other.

$$\text{Hence } 1 \text{ Volt} = \frac{1 \text{ Joule}}{1 \text{ Coulomb}}$$

$$1 \text{ V} = 1 \text{ J/C} = 1 \text{ JC}^{-1}$$

35. Property to oppose the flow of electric current is called resistance. Its SI unit is ohms. If 1V potential difference is there and 1A current is flowing then it is said that there is 1 ohm resistance.

$$\text{i. } \therefore R = \frac{\rho l}{A} \left[\begin{array}{l} \text{where, } l = \text{length of wire,} \\ A = \text{area of cross-section of wire} \end{array} \right]$$

$$R' = \frac{\rho l \times 2}{A}$$

$$\therefore R' = 2R$$

i.e. resistance will be doubled, if length of the wire is doubled.

$$\text{ii. } \therefore R = \frac{\rho l}{A} \Rightarrow R = \frac{\rho l}{\pi r^2} \quad [\because A = \pi r^2]$$

$$R' = \frac{\rho l}{\pi (2r)^2} = \frac{\rho l}{\pi r^2} \times \frac{1}{4} = \frac{R}{4}$$

Thus, resistance will decrease by four times, if radius of wire is doubled.

36. Given :-

Resistor, $R_1 = 4 \Omega$

Resistor, $R_2 = 8 \Omega$

Resistor, $R_3 = 8 \Omega$

Potential Difference, $V = 8 \text{ volts}$.

i. Since two 8Ω resistances are in parallel, their effective resistance (R_p) is given by $\frac{1}{R'} = \frac{1}{8} + \frac{1}{8} + \frac{1}{4}$ or $R' = 4\Omega$

ii. The total resistance in the circuit, $R = 4\Omega + R_p = 4\Omega + 4\Omega = 8\Omega$

Current through the electric circuit, $I = \frac{V}{R} = \frac{8}{4} = 2A$

Since 4Ω resistor and R_p are in series, the current through 4Ω resistors = 1A.

iii. The potential difference across 4Ω resistors, $V = IR = 1 \times 4 = 4V$

iv. Power dissipated in 4Ω resistors, $P = I^2 R = (1A)^2 (4\Omega) = 4W$

v. There is no difference in the readings of ammeters A_1 and A_2 as the same current flows through all elements in a series circuit.

37. a. Relation between resistance and electrical resistivity of the material of conductor in the shape of a cylinder of length and area of cross-section is

$$R = \frac{\rho l}{A}$$

Here, R = resistance, ρ = electrical resistivity of material, l = length and A = area of cross-section

Thus, electrical resistivity, $\rho = \frac{RA}{l}$ (i)

Putting the units of R , A and l in equation (i), we get,

$$\rho = \frac{\Omega \times m^2}{m}$$

$$\rho = \Omega \times m$$

Thus, SI unit of resistivity (ρ) is $\Omega \text{ m}$

b. Given that,

Resistance, $R = 100 \text{ ohm}$, Length, $l = 5 \text{ m}$, Area $A = 3 \times 10^{-7} \text{ m}^2$

The resistivity of the metal is,

$$\rho = \frac{RA}{l}$$

$$\rho = \frac{100 \times 3 \times 10^{-7}}{5}$$

$$\rho = 6 \times 10^{-6} \Omega \text{ m}$$

Hence, the resistivity of the metal is $\rho = 6 \times 10^{-6} \Omega \text{ m}$

38. i. When connected in series $R_s = 3 + 3 + 3 = 9 \text{ ohm}$

ii. When connected in parallel, $\frac{1}{R_p} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{1+1+1}{3} = \frac{3}{3} = 1$ or $R_p = 1 \text{ ohm}$

iii. Two are connected in parallel and third in series with combination.

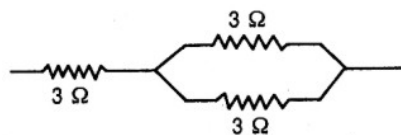
Combined resistance of two resistances of 3 ohm in parallel is

$$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{3} = \frac{1+1}{3} = \frac{2}{3}$$

$$\text{or } R_p = \frac{3}{2} = 1.5 \Omega$$

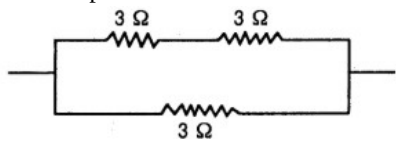
This resistance of 1.5Ω is in series with resistance of 3Ω is

$$R_{\text{total}} = 1.5 + 3 = 4.5\Omega$$



Hence, connected two in parallel and third in series with the combination to get 4.5Ω

Another possible combination is two resistors in series and third in parallel to the combination is in fig.



Two resistors of 3Ω in series becomes $3 + 3 = 6\Omega$

This resistor of 6Ω in parallel with a resistor of 3Ω .

The total resistance in this case is

$$\frac{1}{R'_{Total}} = \frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} = \frac{1}{2}$$

or $R'_{total} = 2\Omega$

Thus two resistors of 3Ω each in series connected to a third resistor of 3Ω in parallel to combination gives 2Ω resistance.

39. a. Power is the rate of doing work. S.I. unit of power = watt.

1 watt = Commercial unit of power kWh.

1 kWh = 3.6×10^6 Joules.

- b. Power consumed by :

2 bulbs = $2 \times 50 = 100W$

Total power consumed per day = $6 \times 100W = 600W = 0.6KWhr$

Power of geyser = $1KW$

Total power consumed per day = $1 \times 1 = 1kwhr$

Total power consumed in 30 days = $1 + 0.6 \times 30 = 48 kwhr$

Cost of electricity = $P \times t = 48 \times 8 = ₹ 384$

40. i. We know that

$$P = \frac{V^2}{R}$$

$$\text{Therefore, } R = \frac{V^2}{P}$$

Resistance of 1st lamp,

$$R_1 = \frac{V^2}{P} = \frac{220 \times 220}{100}$$

$$= 484\Omega$$

Resistance of 2nd lamp,

$$R_2 = \frac{220 \times 220}{100}$$

$$= 4840\Omega$$

Since, two lamps are connected in parallel, so its equivalent resistance is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{484} + \frac{1}{4840}$$

$$R = 440\Omega$$

By Ohm's Law, current drawn from the mains:

$$V = IR$$

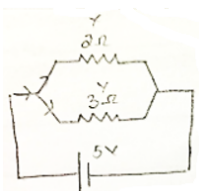
$$220 = I \times 440$$

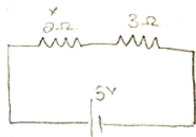
$$I = \frac{220}{440}$$

$$= 0.5 A$$

∴ The current drawn from the mains is $0.5 A$

- ii. a.





b. Voltage across the 3Ω resistor.

According to ohm's law, $V \propto I$

$$V = IR$$

$$I = \frac{V}{R} = \frac{3}{5} \text{ A}$$

$$V = IR$$

$$= \frac{3}{5} \times 5 = 3 \text{ V}$$

41. i. Equivalent resistance between M and N

$$= \left[\frac{(R_3 \times R_4)}{(R_3 + R_4)} \right]$$

ii. Joule's law of heating states that when a current i passes through a conductor of resistance r for time t then the heat developed in the conductor is equal to the product of the square of the current, the resistance, and time. This can be expressed as:

$$H = I^2 R t$$

iii. We need a fuse of 5A for an electric iron which consumes 1 kW power at 220 V. It is because:

$$\text{Given } P = 1000 \text{ W, } V = 220 \text{ V}$$

$$\text{As we know, } P = V \times I$$

$$\text{Or } I = \frac{P}{V}$$

$$\text{Or } I = \frac{1000}{220 \text{ V}} = 4.5 \text{ A}$$

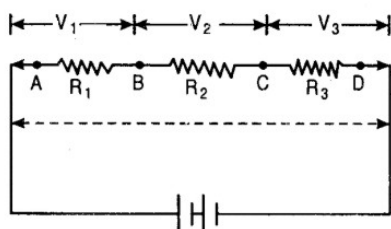
Hence, 4.54 ampere current flows in the circuit, the fuse should be of 5A.

iv. As you know, in a series circuit the current is constant throughout the electric circuit. Thus it is obviously impracticable to connect an electric bulb and an electric heater in series because they need currents of widely different values to operate properly.

42. Combination of resistors: Resistors of all values of resistances are not available. Hence resistors are connected in a number of ways to increase or decrease the combined resistance. There are two distinct ways in which resistors can be connected. They are resistors in series and resistors in parallel.

Resistors connected in series

Resistors are said to be in series if they are joined end to end so that the same current flows through each one of them in succession. Since there is a single path for the moving charge, the same current must flow through each resistor. Let the conductor AB, BC and CD having resistances equal to R_1 , R_2 and R_3 respectively, be joined in series and let the current passing through them be I .



Let V_1 , V_2 and V_3 be the potential difference between the ends of the first, second and third conductor respectively.

By Ohm's law,

$$V_1 = IR_1; V_2 = IR_2 \text{ and } V_3 = IR_3$$

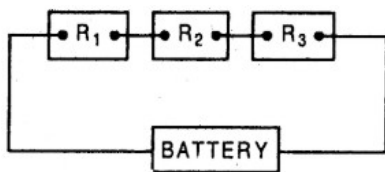
If V is the total potential difference between the ends A and D and R_s is the effective resistance of the combination of all the resistors, then

$$V = IR_s \text{ But } V = V_1 + V_2 + V_3$$

$$\text{OR } IR_s = IR_1 + IR_2 + IR_3$$

$$\text{OR } R_s = R_1 + R_2 + R_3$$

The above result holds good for any number of resistors joined in series.



Thus when some resistors are joined in series, the total resistance is the sum of individual resistances.

43. Resistance of a conductor depends upon the nature of conductor, its temperature and its dimensions. (length, area etc.)

It has been experimentally verified that when temperature, pressure etc. are kept constant, the resistance R of a wire is directly proportional to its length l and inversely proportional to its area A i.e.

$$R \propto l \text{ or } R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

where ρ is a constant called specific resistance or resistivity.

It depends on the nature of the material and its temperature. It does not depend upon its dimensions.

$$\rho = \frac{RA}{l}, A = 1, l = 1, \rho = \frac{R \times 1}{1}$$

$\therefore$ Specific resistance or resistivity is the resistance of a wire of unit length and of unit area cross-section. In other words, the resistivity is the resistance of a unit cube of the wire.

$$\text{SI unit } \rho = \frac{RA}{l} = \frac{\text{ohm m}^2}{\text{m}}$$

$\therefore$ Unit of ρ is ohm m

44. i. It means that if a charge of one C is moved from one point to the other, then one J of work is done. The potential difference is measured by an instrument called the voltmeter.

ii. The electric power is given by

$$P = I^2 R$$

The resistance of the heating element is very high, resistance of connecting cord is very low. Thus, negligible heat generates in the connecting cord and it does not glow.

iii. Based on the resistivity, reason can be explained given below :

a. Silver is a better conductor due to its lower resistivity.

b. Nichrome should be used in electrical heating devices due to very high resistivity.

45. We know that two resistances are in parallel and hence

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\Rightarrow R_p = \frac{R_1 R_2}{R_1 + R_2}$$

$$\text{Given, } R_p = 2\Omega$$

$$\Rightarrow 2 = \frac{R_1 R_2}{R_1 + R_2}$$

$$\Rightarrow 2(R_1 + R_2) = R_1 R_2 \text{ -----(1)}$$

Now, same resistances are in series

$$R_s = R_1 + R_2$$

$$\text{Given, } R_s = 9\Omega$$

$$\Rightarrow R_1 + R_2 = 9\Omega \text{ -----(2)}$$

From (1) and (2), we get

$$R_1 R_2 = 18$$

Again using (2), we have

$$\Rightarrow R_2 = 9 - R_1$$

$$\Rightarrow R_1(9 - R_1) = 18$$

$$R_1^2 - 9R_1 + 18 = 0$$

$$\Rightarrow R_1^2 - 6R_1 - 3R_1 + 18 = 0$$

$$\Rightarrow (R_1 - 6)(R_1 - 3) = 0$$

Either $R_1 = 6$ or $R_1 = 3$

$$R_2 = 3\Omega \text{ or } R_2 = 6\Omega$$

Thus, two resistances are 3Ω and 6Ω

46. (a) The work done by a source of electricity to maintain current in a circuit is known as electrical energy. Its S.I. unit is joule.

(b) (i) Electricity consumed by refrigerator in one day

$$= \text{power} \times \text{time}$$

$$= 400\text{W} \times 10\text{h}$$

$$= 4000\text{Wh}$$

$$= 4\text{kWh}$$

(ii) Electricity consumed by 2 electric fans in 1 day

$$= \text{power} \times \text{time}$$

$$= 2 \times 80\text{W} \times 12\text{h}$$

$$= 1920\text{Wh}$$

$$= 1.92\text{kWh}$$

(iii) Electricity consumed by 6 electric tubes in 1 day

$$= 6 \times 18\text{W} \times 6\text{h}$$

$$= 648\text{Wh} = 0.648 \text{ kWh}$$

Total energy consumed in one day

$$= 4 + 1.92 + 0.648 = 6.548 \text{ kWh}$$

Total energy consumed in one month

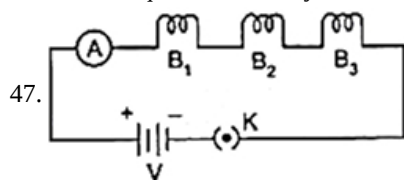
$$= 6.548 \times 30$$

$$= 197.04 \text{ kWh}$$

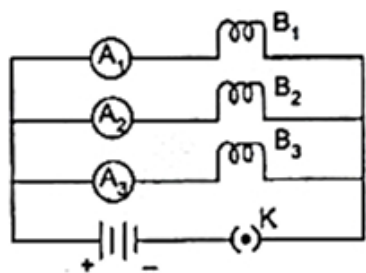
Cost of 1 unit (kWh) = ₹3.00

$$\therefore \text{Cost of } 197.04 \text{ kWh} = 197.04 \times 3$$

$$\therefore \text{The required electricity bill} = ₹591.12$$



Bulbs in series combination



Bulbs in parallel combination

Let the resistance of each bulb is R

In series:

$$\text{Current in each bulb} = \frac{V}{3R}$$

In Parallel:

$$\text{Current in each bulb} = \frac{3V}{R}$$

$$\text{Current will get equally divided in three bulb} = \frac{I}{3} = \frac{V}{R}$$

i. In series, potential difference get divided whereas in parallel potential difference across each bulb remains full. Hence in series bulbs glow with less brightness than bulbs in parallel connection.

ii. Now, if one bulb gets fused in a series connection, all the other bulbs will stop glowing, whereas in a parallel connection, other bulbs will continue to glow. This is because in series combination even a single faulty component break the circuit whereas this does not happen in parallel circuit.

48.	Series Combination	Parallel Combination
	The current has a single path for its flow, hence, same current flows through each resistor.	The main current from the source divides itself in different arms. The current in each resistors is inversely proportional to its resistance.
	The potential difference across the entire circuit is equal to the sum of the potential difference across the individual resistor.	The potential difference across each resistor is same and it is equal to the potential difference across the terminals of the battery (or source).
	The equivalent resistance in series combination is greater	The equivalent resistance in parallel combination is less than the

than the highest resistance in the series combination.	least resistance in the parallel combination.
If one component breaks down, the whole circuit will burn out.	Other components will function even if one component breaks down, each has its own independent circuit

49. a. Power = $VI = \frac{V^2}{R}$, Where V is voltage, I is current, R is resistance.

First case,

$$\text{Power} = 880\text{W} \Rightarrow 880 = 220 \times I \Rightarrow I = 4\text{A}$$

$$\Rightarrow 880 = 220 \times R \Rightarrow R = 55\Omega$$

Second Case,

$$\text{Power} = 440\text{W} = 220 \times I \Rightarrow I = 2\text{A}$$

$$\Rightarrow 440 = 220 \times R \Rightarrow R = 110\Omega$$

- b. When an electric current passes through a conductor (like a high resistance wire) the conductor becomes hot after some time and produces heat. This is called heating effect of Electric Current.

- c. When an electric charge Q moves against a p.d. V, the amount of work done is given by

$$W = Q \times V \dots(i)$$

$$\text{We know, current, } I = \frac{Q}{T}$$

$$Q = I \times T \dots(ii)$$

$$\text{By ohm's law, } \frac{V}{I} = R$$

$$V = I \times R \dots(iii)$$

Putting eqs. (ii) and (iii) in eq. (i),

$$W = I \times t \times I \times R$$

$$W = I^2 R t$$

Assuming that all the electrical work done is converted into heat energy, we get

$$\text{Heat produced, } H = I^2 R t \text{ Jules}$$

This relation is known as Joule's law of heating.

50. The power dissipated in the resistor will increase by 125 percent.

Justification using mathematical calculations.

$$P = I^2 R t$$

$$\text{When I is increased by 50\%, } I' = \frac{3}{2} I.$$

$$\text{Hence, } P' = I'^2 R t = \left(\frac{3}{2} I\right)^2 R t$$

$$P' = \frac{9}{4} I^2 R t$$

$$\therefore \text{Increase in power dissipation} = \frac{9}{4} I^2 R t - I^2 R t = \frac{5}{4} I^2 R t$$

$$\therefore \text{Percentage increase} = \frac{\text{Increase}}{\text{Original}} \times 100\% = \frac{\frac{5}{4} I^2 R t}{I^2 R t} \times 100\% = 125\%$$

51. i. a. A fuse is a safety device used in domestic electric circuits to prevent damages from short circuiting or overloading.

b. An alloy / metal of appropriate (lower) melting point / aluminium / copper / iron / lead etc.

c. In series

ii. 1A, 2A, 3A, 5A, 10A

- iii. a. To protect the circuits and appliances by stopping the flow of unduly high electric current.

b. If current larger than the specified / rated value flows through the circuit, the temperature of the fuse wire increases, this melts the fuse wire and breaks the circuit.

- iv. Power = 1 kW; V = 220 V; I = ?

Formula used

$$P = VI$$

$$\Rightarrow I = \frac{P}{V} = \frac{1,000}{220} = 4.54 \text{ A}$$

Rating of fuse will be 5A and above.

52. In series combination, the second end to each resistance is connected to the first end of the next resistance and so on. The same current flows in all the resistances but the potential difference across each of the resistance is different.

Let three resistance R_1, R_2, R_3 are connected in series with a cell of potential difference V_1, V_2, V_3

According to Ohm's law, we have

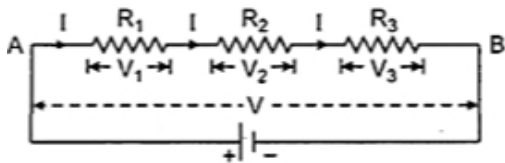
$$V_1 = IR_1, V_2 = IR_2, V_3 = IR_3$$

If the total potential difference between A and B is V, then

$$V = V_1 + V_2 + V_3$$

$$= IR_1 + IR_2 + IR_3$$

$$= I(R_1 + R_2 + R_3)$$



Let the equivalent resistance be R, then

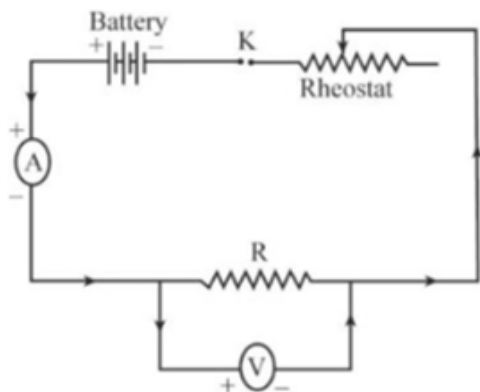
$$V = IR$$

$$\text{and hence } IR = I(R_1 + R_2 + R_3)$$

$$\Rightarrow R = R_1 + R_2 + R_3$$

Therefore, in a series combination of resistance equivalent resistance is equal to the sum of all individual resistances.

53. i.



ii. $V \propto I$ or $\frac{V}{I} = \text{constant}$

For 1st wire

$$R_1 = \rho \frac{1}{A} = 4 \Omega$$

For 2nd wire,

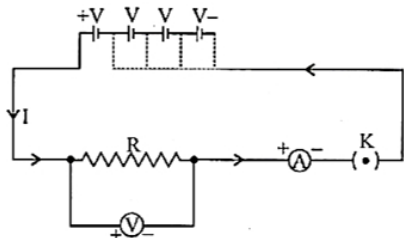
$$R_2 = \rho \frac{1/2}{2A} = \frac{1}{4} \rho \frac{1}{A}$$

$$R_2 = \frac{1}{4} R_1$$

$$R_2 = 1 \Omega$$

54. i. $V = IR$, where V is potential difference, I is current and R is constant for a given resistor as its resistance. Hence, current I is directly proportional to V.

Labelled circuit diagram to verify this relationship is as follows;



ii. An ammeter has a low resistance, which makes it easy for all the current to flow through the circuit without energy loss from heat losses. This means that it does not alter the flow of electric current.

iii. We know that $V = IR$ or $\frac{1}{R} = \frac{1}{V} = \text{slope of graph given.}$

The resistance increases as the slope decreases. Graph B clearly has a greater slope than graph A. As a result, B has less resistance than A. As far as we are aware, combined series resistance is consistently greater than combined parallel resistance. As a result, A and B are series and parallel combinations, respectively.

55. The arrangement of resistance is shown in fig. (a) and its equivalent circuit is shown in fig (b).

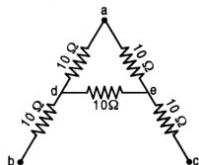


Fig. (a)

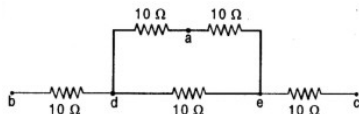


Fig. (b)

Since the wire has a resistance of $1\Omega\text{cm}^{-1}$, the resistance of cross-piece de is 10Ω . Since apex angle is 60° and the cross-piece is 10 cm . long, the ends d and e of the cross-piece must be the mid-point of the legs ab and ac so that ade becomes an equilateral triangles.

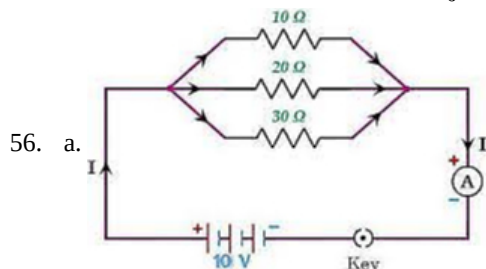
As it is clear from the fig. da and ae are in series. Therefore, resistance of the combination of da and $ae = 10 + 10 = 20\Omega$. This 20Ω resistance is parallel to de .

$\therefore$ Effective resistance r of the portion $daed$ is given by

$$\frac{1}{r} = \frac{1}{10} + \frac{1}{20} = \frac{2+1}{20} = \frac{3}{20} \text{ or } r = \frac{20}{3}$$

Now bd and ec are in series with the resistance of portion $daed$.

$$\therefore \text{Resistance between } b \text{ and } c = 10 + \frac{20}{3} + 10 = \frac{80}{3} = 26.67\Omega$$



56. a.

b. Given : $R_1 = 10\Omega$, $R_2 = 20\Omega$ and $R_3 = 30\Omega$

$$\text{Equivalent resistance of parallel combination} = \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\therefore \frac{1}{R_{eq}} = \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

$$\Rightarrow R_{eq} = 5.45\Omega$$

c. Electric current drawn

$$I = \frac{V}{R} \\ = \frac{10}{5.45} = \frac{0}{45} \text{ ampere}$$

57. a. i. In series: $R_s = R_1 + R_2 + R_3$

$$\text{ii. In parallel: } \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\text{b. Minimum Resistance, } \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{12} = \frac{2}{12} = \frac{1}{6}$$

$$\text{Thus, } R_p = 6\Omega$$

$$\text{Maximum Resistance, } R_s = R_1 + R_2 = 12 + 12 = 24\Omega$$

$$\text{Now, } P = \frac{V^2}{R}$$

$$\frac{P_{\min}}{P_{\max}} = \frac{V^2/R_{\min}}{V^2/R_{\max}} = \frac{R_{\max}}{R_{\min}} = \frac{24}{6} = \frac{4}{1}$$

58. When an electric charge Q moves against a potential difference V , then the amount of work done is given by

$$W = Q \times V \text{ -----(i)}$$

We also know that,

$$Q = I \times t \text{ -----(ii)}$$

and from Ohm's law,

$$V = IR \text{ -----(iii)}$$

Putting the values of Eqs. (ii) and (iii) in Eq. (i), we get

$$W = I \times t \times I \times R$$

$$\therefore \text{Work done, } W = I^2 R t$$

Assuming that all the electrical work done is converted into heat energy.

$$\therefore W = H = I^2 R t$$

Heating effect is desirable because it is useful for the functioning of electrical bulbs, and undesirable because it leads to unnecessary loss of energy.

59. i. Resistance is the quality of a conductor that causes it to resist the flow of an electric current through it. It is the proportion of the potential difference between ends to the current flowing. The SI unit is ohm (Ω).

ii. Two factors on which the resistance of a rectangular conductor depends are:

I. Length of conductor

II. Area of cross-section

$$\text{iii. I. } R = \frac{\rho l}{A}$$

Where, ρ = electrical resistivity

l = length of the conductor

A = cross-sectional area of the conductor

Hence, if the length is double then

$$\Rightarrow R_1 = \rho \frac{(2l)}{A}$$

$$\therefore R_1 = 2(R)$$

So, if the length of the resistance gets doubled then resistance also gets doubled.

II. Now, when the radius is double then

$$\Rightarrow R_2 = \frac{\rho l}{A}$$

$$\Rightarrow R_2 = \frac{\rho l}{\pi (2r)^2}$$

$$\therefore R_2 = \frac{1}{4} (R)$$

So, if the radius gets doubled then resistance will be $\left(\frac{1}{4}\right)^{\text{th}}$ of initial resistance.

60. i. There will be no effect on the glow of the other two bulbs and will remain the same when B_1 gets fused because the three bulbs are connected in parallel and the glowing of the bulb depends on power and the potential difference and resistance remains same of other two bulbs.

ii. When there are parallel connections:

$$\text{Net resistance will be } \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\text{Since resistance is the same so, } R' = \frac{1}{R_3}$$

Applying ohm's law $V = IR$

$$R = 4.5\Omega$$

Since B_2 gets fused, so now only two bulbs B_1 and B_3 are in parallel

$$\text{Therefore net resistance in parallel } \frac{1}{R'} = \frac{2}{R}$$

$$R' = \frac{4.5}{2}\Omega$$

$$I = \frac{V}{R'} = 2 \times \frac{4.5}{4.5}$$

$$I = 2A$$

So, current will be distributed in both the bulbs as 1 A each.

iii. The power dissipated when all three bulbs glow together is

$$P = V \times I$$

$$P = 4.5 \times 3 = 13.5 \text{ W}$$

61. i. Power supplied to the circuit

$$= 1.2 \times 1000 \text{ W} + 200 \text{ W} + 60 \text{ W} + 60 \text{ W}$$

$$= 1200 \text{ W} + 200 \text{ W} + 60 \text{ W} + 60 \text{ W}$$

$$= 1520 \text{ W}$$

$$= 1.52 \text{ kW}$$

$\therefore$ The power supplied to the circuit is 1.52kW

ii. a. Current in the refrigerator = $\frac{\text{Power}}{\text{Voltage}}$ ($\because P = V \times I$)

$$= \frac{200W}{240V}$$

$$= 0.83 \text{ A}$$

∴ The current through the refrigerator is 0.83 A

b. Energy = Power × Time

$$= 1.2 \text{ kW} \times 3\text{h}$$

$$= 1.2 \times 1000 \times 3 \times 60 \times 60\text{s}$$

$$= 1200 \times 3 \times 3600 \text{ J}$$

$$= 12960000 \text{ J}$$

$$= 1.3 \times 10^7 \text{ J}$$

∴ The energy used by the fan in 3 hrs is $1.3 \times 10^7 \text{ J}$

c. Current, $I = \frac{P}{V}$

$$= \frac{60}{240}$$

$$= 0.25 \text{ A}$$

$$\text{Resistance (Filament)} = \frac{V}{I}$$

$$= \frac{240}{0.25}$$

$$= 960 \Omega$$

∴ The resistance of the filament of the bulb is 960Ω

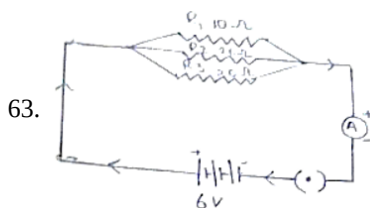
62. It is the potential difference between the ends of the wire which makes the electric charges (or current) to flow in the wire. The potential difference between two points is said to be 1 volt if 1 joule of work is done in moving 1 coulomb of electric charge from one point to another.

$$\text{Thus, } 1 \text{ volt} = \frac{1 \text{ joule}}{1 \text{ coulomb}}$$

$$\text{or } 1\text{V} = \frac{1\text{J}}{1\text{C}} \text{ or } 1\text{JC}^{-1}$$

Given : $W = 12 \text{ J}$, $Q = 2 \text{ C}$

$$\text{Now, potential difference between two terminal } V = \frac{W}{Q} = \frac{12}{2} = 6\text{volts}$$



a. Current through each resistor.

through R_1 , $R_1 = 10 \Omega$, $V = 6\text{V}$

$$V = IR$$

$$I = \frac{V}{R} = \frac{6}{10} = 0.6\text{A}$$

through R_2 , $R_2 = 10\Omega$, $V = 6\text{V}$

$$V = IR$$

$$I = \frac{V}{R} = \frac{6}{10} = 0.3\text{A}$$

through R_3 , $R_3 = 30\Omega$, $V = 6\text{V}$

$$I = \frac{V}{R} = \frac{6}{30} = 0.2\text{A}$$

b. Total current, $I = I_1 + I_2 + I_3$

$$= 0.6 + 0.3 + 0.2 = 1.1\text{A}$$

c. Total Resistance is parallel

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

$$\frac{1}{R_p} = \frac{6+3+2}{60}$$

$$\frac{1}{R_p} = \frac{11}{60}$$

$$(\text{Current Rexiprocol}) = R_p = \frac{60}{11} = 5.4\Omega$$

Section D

64. i. (d) 30Ω

ii. (d) all of these

iii. (a) resistance decreases

- iv. (b) $10^{-8} \Omega\text{m}$ to $10^{-6} \Omega\text{m}$
 v. (b) Alloys does not oxidise readily at high temperature
65. i. (d) all of these
 ii. (b) $H = I^2RT$
 iii. (a) 3380°C
 iv. (c) both (a) and (b)
 v. (a) series
66. i. (a) $V_1 + V_2 + V_3$
 ii. (a) Same in every point of circuit
 iii. (a) $W = VIT$
 iv. (a) 1 ohm
 v. (d) $\frac{15}{2} \Omega$
67. i. (a) $V = \frac{W}{Q}$
 ii. (d) double
 iii. (d) 1 V
 iv. (c) voltage
 v. (a) Alessandra volta
68. i. (c) - Watt
 ii. (b) - ₹ 24
 iii. (d) - all of these
 iv. (a) - 1000 watts
 v. (b) - 110 W
69. i. In series combination, resistance is maximum and in parallel combination, resistance is minimum. according to graph resistance is in parallel as resistance of conductor A is more and current through it is less and also it is obvious from the graph as different current flowing through each resistor which is possible in parallel combination of resistors
 ii. $R_1 = r_1 + r_2 \dots(i)$
 $R_2 = \frac{r_1 r_2}{r_1 + r_2} \dots(ii)$
 $\therefore \frac{R_1}{R_2} = \frac{(r_1 + r_2)^2}{r_1 r_2}$
 iii. In the given circuit, 3 Ω resistors are in series.
 $R_s = 3 + 3 = 6 \Omega$
 Now, R_s and 6 Ω are parallel. $\frac{1}{R_p} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} \Rightarrow R_p = 3\Omega$
- OR**
- $V = 0.2 \times 10 = 2 \text{ V}$
 So, total voltage supplied is same as 2 V.
70. i. $E \propto t$
 ii. Given: $P = 60 \text{ W}$, $t = 1 \text{ min}$
 $E = 60 \times 1 \times 60 = 3600 \text{ J}$
 iii. Given: $P = 400 \Omega$, $t = 8 \text{ hour}$
 $E = 400 \times 8 = 3200 \text{ Wh} = 3.2 \text{ kWh}$
 $\text{Cost} = 3.2 \times 5 = ₹16$
 iv. Given: $I = 5\text{A}$, $R = 2\Omega$, $t = 30 \text{ min}$
 $E = I^2 R t = 5 \times 5 \times 2 \times 30 \times 60$
 $E = 90000 \text{ J} = 90 \text{ kJ}$
71. i. $q = 2 \text{ C}$, $t = 100 \text{ ms} = 0.1 \text{ s}$
 $I = \frac{q}{t} = \frac{2}{0.1} = 20 \text{ A}$
 ii. $W = 100 \text{ J}$, $q = 20 \text{ C}$
 $V = \frac{W}{q} = \frac{100}{20} = 5 \text{ V}$
 iii. $I = 1 \text{ A}$, $t = 1 \text{ s}$
 $q = It = 1 \times 1 = 1 \text{ C}$
 $n = \frac{q}{e} = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18}$
- OR**

$$V = \frac{W}{q} = \frac{W}{It}$$

72. i. In series combination, $R_s = R_1 + R_2 + R_3 = R + R + R = 3R$

ii. Resistance of each wire = $\frac{20}{4} = 5 \Omega$

Equivalent resistance in series

$$R_s = 5 + 5 + 5 + 5 = 20 \Omega$$

iii. All are in series, $R_s = 5R = 5 \times 2 = 10 \Omega$

OR

$$R_s = 1 + 2 + 3 = 6 \Omega$$

$$I = \frac{18}{6} = 3 \text{ A}$$

73. i. Low resistance, high melting point

ii. High resistance, low melting point

iii. Given: $H = I^2 R t$

$$\text{So, } H' = (2I)^2 \cdot \frac{R}{2} t = 2H$$

iv. Given: $I = 5 \text{ A}$, resistance = R . Let r be the new radius.

$$\text{Now, } H = i^2 R t \dots (a)$$

$$\text{Also } H' = I'^2 R' t \dots (b)$$

$$\text{From (a) and (b), } 5^2 \times \rho \frac{L}{\pi r^2} t = 10^2 \times \rho \frac{L}{\pi r'^2} \cdot t$$

$$\frac{25}{r^2} = \frac{100}{r'^2} \Rightarrow \frac{r'}{r} = 2 \Rightarrow r' = 2r$$

74. i. The equivalent resistance in the parallel combination is lesser than the least value of the individual resistance.

ii. Resistance of each piece = $\frac{12}{3} = 4 \Omega$

$$\frac{1}{R_p} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4} \Rightarrow R_p = \frac{4}{3} \Omega$$

iii. All the three resistors are in parallel.

$$\therefore \frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} + \frac{1}{1} = \frac{1+2+6}{6} = \frac{9}{6} \quad R_p = \frac{6}{9} = \frac{2}{3} \Omega$$

OR

All are in parallel.

$$\frac{1}{R_p} = \frac{1}{12} \times 4 = \frac{1}{3} \Rightarrow R_p = 3 \Omega$$

$$I = \frac{3}{3} = 1 \text{ A}$$

$$\text{So, current in each resistor } I' = \frac{3}{12} = \frac{1}{4} \text{ A}$$

75. i. The electrical wire should have low resistance.

ii. The slope of the given graph gives us the inverse of resistance. Resistance of a material increases with increasing the temperature. Because the collision between the molecules increases. In the graph given, T_A has smaller slope and hence corresponds to higher resistance. Therefore, $T_A > T_B$

iii. $R_1 = \rho \frac{L}{A}, R_2 = \rho \frac{2L}{A/2}$

$$R_1 = R_2$$

OR

$$R' = \frac{R}{5} = \frac{20}{5} = 4 \Omega$$